

# **A Meta-Analysis of the Relationship between Taxes and Economic Growth in the U.S. States**

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University of Canterbury

Professor Bob Reed  
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**OR...**

# **PCC Versus Estimated Coefficients: Will the Real MRA Please Stand Up?**

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## SUBJECT: The Relationship Between Taxes and Economic Growth in US States

Table3.3 Final keywords and key terms

| Seq. | keywords                                                                     |
|------|------------------------------------------------------------------------------|
| 1    | state and local taxes AND state economic growth                              |
| 2    | taxes AND state economic growth                                              |
| 3    | tax structure AND state economic growth                                      |
| 4    | state taxes AND state economic performance                                   |
| 5    | the effect of fiscal policy on state or regional economic growth             |
| 6    | taxation and state economic growth                                           |
| 7    | tax policy and economic growth                                               |
| 8    | taxes and GSP                                                                |
| 9    | taxes and per capita income growth                                           |
| 10   | fiscal policy and economic growth, fiscal policies AND state economic growth |
| 11   | state and local taxes AND state growth                                       |
| 12   | local and state tax AND economic growth                                      |
| 13   | state and local taxes AND economic development                               |
| 14   | taxation AND state economic growth                                           |
| 15   | tax policy and state-level economic growth                                   |
| 16   | fiscal structures and economic growth at the state level                     |
| 17   | tax rates AND state economic growth                                          |
| 18   | tax policy and state economic growth                                         |
| 19   | taxing AND economic growth                                                   |
| 20   | state fiscal policy AND state economic performance/ economic growth          |
| 21   | state taxation AND per capita income growth                                  |

### SEARCH SOURCES:

- EconLit
- Google Scholar
- JSTOR
- RePEc
- SSRN
- Ebsco
- Web of Science
- Scopus
- Etc.

### SEARCH PERIOD:

- February – April, 2014

## OTHER SOURCES:

Table3.4 List of Think Tanks and Research Firms

| <b>Seq.</b> | <b>Name</b>                               | <b>E-mail Address</b>                                                      |
|-------------|-------------------------------------------|----------------------------------------------------------------------------|
| 1           | The Heritage Foundation                   | <a href="http://www.heritage.org">www.heritage.org</a>                     |
| 2           | Economic Policy Institute                 | <a href="http://www.epi.org">www.epi.org</a>                               |
| 3           | Institute on Taxation and Economic Policy | <a href="http://www.itep.org">www.itep.org</a>                             |
| 4           | Tax foundation                            | <a href="http://Taxfoundation.org">Taxfoundation.org</a>                   |
| 5           | Tax policy centre                         | <a href="http://Taxpolicycenter.org">Taxpolicycenter.org</a>               |
| 6           | National tax association                  | <a href="http://Ntanet.org">Ntanet.org</a>                                 |
| 7           | Centre on Budget and policy priorities    | <a href="http://www.cbpp.org">www.cbpp.org</a>                             |
| 8           | Tax analysts                              | <a href="http://www.taxanalysts.com">www.taxanalysts.com</a>               |
| 9           | National centre for policy analysis       | <a href="http://www.ncpa.org">www.ncpa.org</a>                             |
| 10          | Council on State Taxation                 | <a href="http://www.cost.org">www.cost.org</a>                             |
| 11          | Goldwater Institute                       | <a href="http://www.goldwaterinstitute.org">www.goldwaterinstitute.org</a> |
| 12          | U.S. Department of State                  | <a href="http://www.state.gov">www.state.gov</a>                           |
| 13          | Centre of Freedom and Prosperity          | <a href="http://Freedomandprosperity.org">Freedomandprosperity.org</a>     |
| 14          | The Federal Reserve Bank of Minneapolis   | <a href="http://www.minneapolisfed.org">www.minneapolisfed.org</a>         |
| 15          | CATO Institute                            | <a href="http://www.cato.org">www.cato.org</a>                             |
| 16          | Federation of American Scientists         | <a href="http://www.fas.org">www.fas.org</a>                               |

## NUMBER OF STUDIES IDENTIFIED BY SEARCH:

- 400+ studies

## INCLUSION CRITERIA:

- 1) Dependent Variable = A measure of state income growth
- 2) Composition = Must include at least 44 contiguous US states
- 3) Tax Variable = Must be measured as a rate (taxes per unit of income)

## NOTABLE ADDITIONAL SOURCES:

- Contacted other researchers in the field

## FINAL SAMPLE:

27 studies / 939 estimates

# TYPICAL EXAMPLE:

**TABLE 7**  
Estimation of Pooled Regressions Using Disaggregated State Fiscal Data

| Variable              | Model with Disaggregated Tax Rates |                    |                   | Model with Tax Rates, Disaggregated State Public Expenditures, and Federal Aid |                     |                     |
|-----------------------|------------------------------------|--------------------|-------------------|--------------------------------------------------------------------------------|---------------------|---------------------|
|                       | Model (2)                          | Model (3)          | Model (4)         | Model (2)                                                                      | Model (3)           | Model (4)           |
| $\mu$                 | 0.933<br>(0.666)                   | —                  | —                 | 1.086<br>(1.035)                                                               | —                   | —                   |
| <i>INITIAL</i>        | -0.403<br>(0.820)                  | —                  | —                 | -0.0002**<br>(0.0001)                                                          | —                   | —                   |
| <i>t</i>              | 0.045**<br>(0.018)                 | 0.059**<br>(0.024) | —                 | 0.030<br>(0.039)                                                               | 0.042<br>(0.047)    | —                   |
| $\tau_{income}$       | -0.191<br>(0.210)                  | -0.264<br>(0.564)  | 0.104<br>(0.497)  | 0.146<br>(0.213)                                                               | 0.785<br>(0.487)    | 0.290<br>(0.401)    |
| $\tau_{corporate}$    | 1.725*<br>(0.868)                  | 6.090*<br>(3.104)  | 3.821<br>(2.823)  | 2.841**<br>(0.974)                                                             | 6.986**<br>(2.273)  | 7.215**<br>(2.235)  |
| $\tau_{sales}$        | -0.028<br>(1.527)                  | 6.607<br>(5.606)   | 9.581<br>(6.978)  | 2.264<br>(1.765)                                                               | 3.190<br>(4.379)    | 5.647<br>(4.031)    |
| $\tau_{property}$     | -1.595<br>(1.790)                  | -1.951<br>(1.920)  | -0.797<br>(1.535) | 0.133<br>(0.551)                                                               | 1.656<br>(1.617)    | 1.977<br>(1.531)    |
| <i>EDUCATION</i>      | —                                  | —                  | —                 | -0.511*<br>(0.158)                                                             | -0.338<br>(0.292)   | -0.490<br>(0.779)   |
| <i>WELFARE</i>        | —                                  | —                  | —                 | -0.908**<br>(0.200)                                                            | -1.094**<br>(0.299) | -1.156**<br>(0.323) |
| <i>HOSPITAL</i>       | —                                  | —                  | —                 | 0.686<br>(0.622)                                                               | -0.207<br>(0.931)   | -1.073<br>(1.023)   |
| <i>HIGHWAYS</i>       | —                                  | —                  | —                 | -0.366<br>(0.223)                                                              | 0.065<br>(0.289)    | 0.023<br>(0.283)    |
| <i>Government aid</i> | —                                  | —                  | —                 | 0.224<br>(0.152)                                                               | 0.324<br>(0.380)    | 0.399<br>(0.343)    |
| Adjusted $R^2$        | 0.226                              | 0.215              | 0.505             | 0.252                                                                          | 0.255               | 0.475               |

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|                       | Model (2)                          | Model (3)          | Model (4)         | Model (2)                                                                      | Model (3)           | Model (4)           |
| $\mu$                 | 0.933<br>(0.666)                   | —                  | —                 | 1.086<br>(1.035)                                                               | —                   | —                   |
| <i>INITIAL</i>        | -0.403<br>(0.820)                  | —                  | —                 | -0.0002**<br>(0.0001)                                                          | —                   | —                   |
| <i>t</i>              | 0.045**<br>(0.018)                 | 0.059**<br>(0.024) | —                 | 0.030<br>(0.039)                                                               | 0.042<br>(0.047)    | —                   |
| $\tau_{income}$       | -0.191<br>(0.210)                  | -0.264<br>(0.564)  | 0.104<br>(0.497)  | 0.146<br>(0.213)                                                               | 0.785<br>(0.487)    | 0.290<br>(0.401)    |
| $\tau_{corporate}$    | 1.725*<br>(0.868)                  | 6.090*<br>(3.104)  | 3.821<br>(2.823)  | 2.841**<br>(0.974)                                                             | 6.986**<br>(2.273)  | 7.215**<br>(2.235)  |
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| $\tau_{property}$     | -1.595<br>(1.790)                  | -1.951<br>(1.920)  | -0.797<br>(1.535) | 0.133<br>(0.551)                                                               | 1.656<br>(1.617)    | 1.977<br>(1.531)    |
| <i>EDUCATION</i>      | —                                  | —                  | —                 | -0.511*<br>(0.158)                                                             | -0.338<br>(0.292)   | -0.490<br>(0.779)   |
| <i>WELFARE</i>        | —                                  | —                  | —                 | -0.908**<br>(0.200)                                                            | -1.094**<br>(0.299) | -1.156**<br>(0.323) |
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| <i>Government aid</i> | —                                  | —                  | —                 | 0.224<br>(0.152)                                                               | 0.324<br>(0.380)    | 0.399<br>(0.343)    |
| Adjusted $R^2$        | 0.226                              | 0.215              | 0.505             | 0.252                                                                          | 0.255               | 0.475               |

## CODING OF TAX VARIABLES

### ***VARIABLE GROUP = tax measures (Part 1)***

|                    |                                                                       |
|--------------------|-----------------------------------------------------------------------|
| <i>PI-Tax</i>      | =1, if personal income tax is used.                                   |
| <i>CO-Tax</i>      | =1, if corporate income tax is used.                                  |
| <i>SA-Tax</i>      | =1, if sales tax is used.                                             |
| <i>PR-Tax</i>      | =1, if property tax is used.                                          |
| <i>OT-Tax</i>      | =1, if other taxes are used.                                          |
| <i>Overall tax</i> | =1, if the overall tax (no decomposition) is used. (Omitted category) |

## CODING OF TAX VARIABLES (continued)

### ***VARIABLE GROUP = tax measures (Part 2)***

|                    |                                                                                                 |
|--------------------|-------------------------------------------------------------------------------------------------|
| <i>Marginal</i>    | =1, if the marginal form of tax is used.                                                        |
| <i>Differenced</i> | =1, if the differenced form of tax is used.                                                     |
| <i>Num-Tax</i>     | The number of tax variables in the estimation.                                                  |
| <i>Effective</i>   | =1, if the tax variable in estimation is effective tax rate as opposed to a statutory tax rate. |

## CODING OF TAX VARIABLES (continued)

### ***VARIABLE GROUP = tax measures (Part 3)***

|                              |                                                    |
|------------------------------|----------------------------------------------------|
| <i>First-Lag</i>             | =1, if the first lagged form of tax is used.       |
| <i>More-Lag</i>              | =1, if the second/more lagged form of tax is used. |
| <i>No Lag-<br/>(current)</i> | =1, if no lag is used.(Omitted category)           |

## A COMPLICATION:

### The Government Budget Constraint

$$\sum_{e=1}^E Expenditures_e + \sum_{r=1}^R Revenues_r = Deficit$$

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| $\tau_{corporate}$    | 1.725*<br>(0.868)                  | 6.090*<br>(3.104)  | 3.821<br>(2.823)  | 2.841**<br>(0.974)                                                             | 6.986**<br>(2.273)  | 7.215**<br>(2.235)  |
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| <i>WELFARE</i>        | —                                  | —                  | —                 | -0.908**<br>(0.200)                                                            | -1.094**<br>(0.299) | -1.156**<br>(0.323) |
| <i>HOSPITAL</i>       | —                                  | —                  | —                 | 0.686<br>(0.622)                                                               | -0.207<br>(0.931)   | -1.073<br>(1.023)   |
| <i>HIGHWAYS</i>       | —                                  | —                  | —                 | -0.366<br>(0.223)                                                              | 0.065<br>(0.289)    | 0.023<br>(0.283)    |
| <i>Government aid</i> | —                                  | —                  | —                 | 0.224<br>(0.152)                                                               | 0.324<br>(0.380)    | 0.399<br>(0.343)    |
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| <i>WELFARE</i>        | —                                  | —                  | —                 | -0.908**<br>(0.200)                                                            | -1.094**<br>(0.299) | -1.156**<br>(0.323) |
| <i>HOSPITAL</i>       | —                                  | —                  | —                 | 0.686<br>(0.622)                                                               | -0.207<br>(0.931)   | -1.073<br>(1.023)   |
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## A COMPLICATION:

### The Government Budget Constraint

$$\sum_{e=1}^E Expenditures_e + \sum_{r=1}^R Revenues_r = Deficit$$

Table 1

## Theoretical aggregation of functional classifications

| Theoretical classification | Functional classification                                                                                                                                                     |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Distortionary taxation     | Taxation on income and profit<br>Social security contributions<br>Taxation on payroll and manpower<br>Taxation on property                                                    |
| Non-distortionary taxation | Taxation on domestic goods and services                                                                                                                                       |
| Other revenues             | Taxation on international trade<br>Non-tax revenues<br>Other tax revenues                                                                                                     |
| Productive expenditures    | General public services expenditure<br>Defence expenditure<br>Educational expenditure<br>Health expenditure<br>Housing expenditure<br>Transport and communication expenditure |
| Unproductive expenditures  | Social security and welfare expenditure<br>Expenditure on recreation<br>Expenditure on economic services                                                                      |
| Other expenditures         | Other expenditure (unclassified)                                                                                                                                              |

Note: functional classifications refer to the classifications given in the data source.

*Pred-Neg*

=1, if the theoretical prediction of the coefficient is negative.

| Theoretical classification | Functional classification                                                                                                                                                     |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Distortionary taxation     | Taxation on income and profit<br>Social security contributions<br>Taxation on payroll and manpower<br>Taxation on property                                                    |
| Non-distortionary taxation | Taxation on domestic goods and services                                                                                                                                       |
| Other revenues             | Taxation on international trade<br>Non-tax revenues<br>Other tax revenues                                                                                                     |
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| Unproductive expenditures  | Social security and welfare expenditure<br>Expenditure on recreation<br>Expenditure on economic services                                                                      |
| Other expenditures         | Other expenditure (unclassified)                                                                                                                                              |

*Pred-Pos*

=1, if the theoretical prediction of the coefficient is positive.

| Theoretical classification | Functional classification                                                                                                                                                     |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Distortionary taxation     | Taxation on income and profit<br>Social security contributions<br>Taxation on payroll and manpower<br>Taxation on property                                                    |
| Non-distortionary taxation | Taxation on domestic goods and services                                                                                                                                       |
| Other revenues             | Taxation on international trade<br>Non-tax revenues<br>Other tax revenues                                                                                                     |
| Productive expenditures    | General public services expenditure<br>Defence expenditure<br>Educational expenditure<br>Health expenditure<br>Housing expenditure<br>Transport and communication expenditure |
| Unproductive expenditures  | Social security and welfare expenditure<br>Expenditure on recreation<br>Expenditure on economic services                                                                      |
| Other expenditures         | Other expenditure (unclassified)                                                                                                                                              |

Briem (1994 Working paper):

PRED-NEG, Business taxes/Corporate income taxes to fund state transfer payments

| Tax                                     | (1) Business Taxes |               | (2) Corporate Taxes only |               |
|-----------------------------------------|--------------------|---------------|--------------------------|---------------|
| Rate:                                   |                    |               |                          |               |
| Expected Growth                         | 0.361 (4.71)       | 0.462 (4.96)  | 0.298 (3.62)             | 0.290 (3.00)  |
| Tax Rate <sub>t-2</sub>                 | 0.010 (0.74)       | -0.025 (1.37) | 0.032 (3.00)             | 0.039 (2.57)  |
| Tax Rate <sub>t-4</sub>                 | -0.040 (2.38)      | 0.002 (0.08)  | -0.034 (1.97)            | -0.025 (0.96) |
| Infrastructure <sub>t-2</sub>           | -0.001 (0.19)      | -0.024 (3.35) | -0.006 (0.88)            | -0.029 (4.10) |
| Infrastructure <sub>t-4</sub>           | -0.156 (4.79)      | -0.210 (5.14) | -0.152 (4.23)            | -0.237 (4.97) |
| Public Capital <sub>t-2</sub>           | 0.053 (2.60)       | 0.084 (2.38)  | 0.067 (3.18)             | 0.0559 (1.59) |
| Public Capital <sub>t-4</sub>           | 0.005 (0.87)       | 0.006 (1.02)  | 0.003 (0.58)             | 0.008 (1.47)  |
| Education <sub>t-2</sub>                | -0.000 (0.10)      | -0.005 (0.99) | 0.000 (0.05)             | -0.008 (1.70) |
| Education <sub>t-4</sub>                | 0.073 (1.63)       | -0.047 (0.78) | 0.054 (1.18)             | 0.126 (1.46)  |
| Higher Education <sub>t-2</sub>         | -0.015 (1.17)      | -0.166 (4.10) | -0.025 (1.87)            | -0.148 (3.59) |
| Higher Education <sub>t-4</sub>         | 0.004 (0.32)       | -0.008 (0.67) | -0.020 (1.94)            | -0.007 (0.71) |
| Electricity <sub>t-2</sub>              | 0.021 (1.28)       | 0.039 (2.34)  | 0.014 (0.85)             | 0.027 (1.66)  |
| Electricity <sub>t-4</sub>              | -0.004 (0.64)      | -0.020 (2.91) | -0.007 (1.12)            | -0.021 (3.19) |
| Industrial Revenue Bonds <sub>t-2</sub> | 0.101 (3.07)       | 0.072 (2.43)  | 0.077 (2.20)             | 0.017 (0.53)  |
| Industrial Revenue Bonds <sub>t-4</sub> | -0.110 (1.66)      | -0.145 (2.32) | -0.117 (1.72)            | -0.145 (2.32) |
| Wage <sub>t-2</sub>                     | -0.017 (2.81)      | -0.011 (1.97) | -0.011 (1.72)            | -0.000 (0.03) |
| Wage <sub>t-4</sub>                     | -0.007 (1.59)      | -0.003 (0.81) | -0.007 (1.54)            | -0.002 (0.52) |
| Unionization <sub>t-2</sub>             | -0.089 (2.07)      | -0.096 (2.31) | -0.053 (1.22)            | -0.086 (2.12) |
| Unionization <sub>t-4</sub>             | 0.002 (0.16)       | 0.013 (1.19)  | 0.008 (0.59)             | 0.014 (1.21)  |

Briem (1994 Working paper):

PRED-NEG, Business taxes/Corporate income taxes to fund state transfer payments

| Tax                                     | (1) Business Taxes |               | (2) Corporate Taxes only |               |
|-----------------------------------------|--------------------|---------------|--------------------------|---------------|
| Rate:                                   |                    |               |                          |               |
| Expected Growth                         | 0.361 (4.71)       | 0.462 (4.96)  | 0.298 (3.62)             | 0.290 (3.00)  |
| Tax Rate <sub>t-2</sub>                 | 0.010 (0.74)       | -0.025 (1.37) | 0.032 (3.00)             | 0.039 (2.57)  |
| Tax Rate <sub>t-4</sub>                 | -0.040 (2.38)      | 0.002 (0.08)  | -0.034 (1.97)            | -0.025 (0.96) |
| Infrastructure <sub>t-2</sub>           | -0.001 (0.19)      | -0.024 (3.33) | -0.006 (0.68)            | -0.029 (4.10) |
| Infrastructure <sub>t-4</sub>           | -0.156 (4.79)      | -0.210 (5.14) | -0.152 (4.23)            | -0.237 (4.97) |
| Public Capital <sub>t-2</sub>           | 0.053 (2.60)       | 0.084 (2.38)  | 0.067 (3.18)             | 0.0559 (1.59) |
| Public Capital <sub>t-4</sub>           | 0.005 (0.87)       | 0.006 (1.02)  | 0.003 (0.58)             | 0.008 (1.47)  |
| Education <sub>t-2</sub>                | -0.000 (0.10)      | -0.005 (0.99) | 0.000 (0.05)             | -0.008 (1.70) |
| Education <sub>t-4</sub>                | 0.073 (1.63)       | -0.047 (0.78) | 0.054 (1.18)             | 0.126 (1.46)  |
| Higher Education <sub>t-2</sub>         | -0.015 (1.17)      | -0.166 (4.10) | -0.025 (1.87)            | -0.148 (3.59) |
| Higher Education <sub>t-4</sub>         | 0.004 (0.32)       | -0.008 (0.67) | -0.020 (1.94)            | -0.007 (0.71) |
| Electricity <sub>t-2</sub>              | 0.021 (1.28)       | 0.039 (2.34)  | 0.014 (0.85)             | 0.027 (1.66)  |
| Electricity <sub>t-4</sub>              | -0.004 (0.64)      | -0.020 (2.91) | -0.007 (1.12)            | -0.021 (3.19) |
| Industrial Revenue Bonds <sub>t-2</sub> | 0.101 (3.07)       | 0.072 (2.43)  | 0.077 (2.20)             | 0.017 (0.53)  |
| Industrial Revenue Bonds <sub>t-4</sub> | -0.110 (1.66)      | -0.145 (2.32) | -0.117 (1.72)            | -0.145 (2.32) |
| Wage <sub>t-2</sub>                     | -0.017 (2.81)      | -0.011 (1.97) | -0.011 (1.72)            | -0.000 (0.03) |
| Wage <sub>t-4</sub>                     | -0.007 (1.59)      | -0.003 (0.81) | -0.007 (1.54)            | -0.002 (0.52) |
| Unionization <sub>t-2</sub>             | -0.089 (2.07)      | -0.096 (2.31) | -0.053 (1.22)            | -0.086 (2.12) |
| Unionization <sub>t-4</sub>             | 0.002 (0.16)       | 0.013 (1.19)  | 0.008 (0.59)             | 0.014 (1.21)  |

Briem (1994 Working paper):

PRED-NEG, Business taxes/Corporate income taxes to fund state transfer payments

| Tax                                     | (1) Business Taxes |               | (2) Corporate Taxes only |               |
|-----------------------------------------|--------------------|---------------|--------------------------|---------------|
| Rate:                                   |                    |               |                          |               |
| Expected Growth                         | 0.361 (4.71)       | 0.462 (4.96)  | 0.298 (3.62)             | 0.290 (3.00)  |
| Tax Rate <sub>t-2</sub>                 | 0.010 (0.74)       | -0.025 (1.37) | 0.032 (3.00)             | 0.039 (2.57)  |
| Tax Rate <sub>t-4</sub>                 | 0.040 (2.38)       | 0.002 (0.08)  | 0.034 (1.07)             | 0.025 (0.96)  |
| Infrastructure <sub>t-2</sub>           | -0.001 (0.19)      | -0.024 (3.35) | -0.006 (0.88)            | -0.029 (4.10) |
| Infrastructure <sub>t-4</sub>           | -0.156 (4.79)      | -0.210 (5.14) | -0.152 (4.23)            | -0.237 (4.97) |
| Public Capital <sub>t-2</sub>           | 0.053 (2.60)       | 0.084 (2.38)  | 0.067 (3.18)             | 0.0559 (1.59) |
| Public Capital <sub>t-4</sub>           | 0.005 (0.87)       | 0.006 (1.02)  | 0.003 (0.58)             | 0.008 (1.47)  |
| Education <sub>t-2</sub>                | -0.000 (0.10)      | -0.005 (0.99) | 0.000 (0.05)             | -0.008 (1.70) |
| Education <sub>t-4</sub>                | 0.073 (1.63)       | -0.047 (0.78) | 0.054 (1.18)             | 0.126 (1.46)  |
| Higher Education <sub>t-2</sub>         | -0.015 (1.17)      | -0.166 (4.10) | -0.025 (1.87)            | -0.148 (3.59) |
| Higher Education <sub>t-4</sub>         | 0.004 (0.32)       | -0.008 (0.67) | -0.020 (1.94)            | -0.007 (0.71) |
| Electricity <sub>t-2</sub>              | 0.021 (1.28)       | 0.039 (2.34)  | 0.014 (0.85)             | 0.027 (1.66)  |
| Electricity <sub>t-4</sub>              | -0.004 (0.64)      | -0.020 (2.91) | -0.007 (1.12)            | -0.021 (3.19) |
| Industrial Revenue Bonds <sub>t-2</sub> | -0.101 (3.07)      | -0.072 (2.45) | -0.077 (2.20)            | -0.077 (0.55) |
| Industrial Revenue Bonds <sub>t-4</sub> | -0.110 (1.66)      | -0.145 (2.32) | -0.117 (1.72)            | -0.145 (2.32) |
| Wage <sub>t-2</sub>                     | -0.017 (2.81)      | -0.011 (1.97) | -0.011 (1.72)            | -0.000 (0.03) |
| Wage <sub>t-4</sub>                     | -0.007 (1.59)      | -0.003 (0.81) | -0.007 (1.54)            | -0.002 (0.52) |
| Unionization <sub>t-2</sub>             | -0.089 (2.07)      | -0.096 (2.31) | -0.053 (1.22)            | -0.086 (2.12) |
| Unionization <sub>t-4</sub>             | 0.002 (0.16)       | 0.013 (1.19)  | 0.008 (0.59)             | 0.014 (1.21)  |

Clark and Miller (2014 working paper):  
 PRED-POS, Sales taxes to fund health and hospital expenditures

**Table 6: Long-Run and Short-Run Effects of Disaggregated Revenues and Expenditures on Growth 1977-2010  
 (continued)**

| Variable                                   | Higher<br>Educ. Exp. | Second.<br>Educ. Exp. | Health &<br>Hosp. Exp. | Parks, Rec., &<br>Nat. Res. Exp | Wel., Unem., Hous.,<br>& Comm. Exp. | Other Exp. | Net Intergov't.<br>Revenue | Surplus |
|--------------------------------------------|----------------------|-----------------------|------------------------|---------------------------------|-------------------------------------|------------|----------------------------|---------|
| Lag Real Person. Inc. Per Capita           | -6.89*               | -6.89*                | -6.89*                 | -6.89*                          | -6.89*                              | -6.89*     | -6.89*                     | -6.89*  |
| Unemployment Rate                          | -0.76*               | -0.76*                | -0.76*                 | -0.76*                          | -0.76*                              | -0.76*     | -0.76*                     | -0.76*  |
| Population Growth Rate                     | -0.56**              | -0.56**               | -0.56**                | -0.56**                         | -0.56**                             | -0.56**    | -0.56**                    | -0.56** |
| Δ Property Tax                             | 0.11                 | -0.96**               | -0.76                  | -7.15‡                          | -0.26                               | -0.44      | 0.71                       | 0.16    |
| Lagged Property Tax.                       | -2.34**              | -0.19                 | -1.35                  | -6.03**                         | 0.27                                | -0.35      | 1.08                       | 0.23    |
| Δ Sales Tax                                | -0.39                | -1.45*                | -1.26                  | -7.65**                         | -0.76                               | -0.93      | 0.21                       | -0.34   |
| Lagged Sales Tax                           | -2.77**              | -0.62                 | -1.78                  | -6.46**                         | -0.15                               | -0.78      | 0.65                       | -0.20   |
| Δ Individual Income Tax                    | 0.17                 | -0.90                 | -0.70                  | -7.09**                         | -0.20                               | -0.38      | 0.77                       | 0.22    |
| Lagged Individual Income Tax               | -1.21                | 0.94                  | -0.22                  | -4.90‡                          | 1.41                                | 0.78       | 2.21**                     | 1.37**  |
| Δ Corporate Income Tax                     | 1.82                 | 0.75                  | 0.94                   | -5.45                           | 1.44                                | 1.27       | 2.42**                     | 1.87    |
| Lagged Corporate Income Tax                | -1.99                | 0.17                  | -0.99                  | -5.68**                         | 0.63                                | 0.01       | 1.43                       | 0.59    |
| Δ Other Revenue                            | 0.76                 | -0.31                 | -0.11                  | -6.50**                         | 0.39                                | 0.21       | 1.36                       | 0.81**  |
| Lagged Other Revenue                       | -2.12**              | 0.03                  | -1.12                  | -5.81**                         | 0.50                                | -0.13      | 1.30‡                      | 0.46‡   |
| Δ Net Utility Expenditure                  | 0.65                 | 1.72**                | 1.52                   | 7.91**                          | 1.02‡                               | 1.20       | 0.05                       | 0.60    |
| Lagged Net Utility Expenditure             | 3.25**               | 1.10                  | 2.26                   | 6.94**                          | 0.63                                | 1.26       | -0.17                      | 0.67    |
| Δ Police, Fire, & Correct. Facilities      | -1.77                | -0.70                 | -0.90                  | 5.49                            | -1.39                               | -1.22      | -2.37**                    | -1.82** |
| Lagged Police, Fire, & Correct. Facilities | 3.09**               | 0.94                  | 2.10                   | 6.78**                          | 0.47                                | 1.10       | -0.33                      | 0.52    |
| Δ Transportation Expenditure               | -1.29                | -0.22                 | -0.42                  | 5.97‡                           | -0.91                               | -0.74      | -1.89                      | -1.34‡  |
| Lagged Transportation Expenditure          | 1.54                 | -0.61                 | 0.55                   | 5.23‡                           | -1.08‡                              | -0.45      | -1.88*                     | -1.04‡  |
| Δ Higher Education Expenditure             |                      | 1.07                  | 0.87                   | 7.26**                          | 0.37                                | 0.55       | -0.60                      | -0.05   |
| Lagged Higher Education Expenditure        |                      | -2.15‡                | -0.99                  | 3.69                            | -2.62**                             | -1.99‡     | -3.42*                     | -2.58** |
| Δ Secondary Education Expenditure.         | -1.07                |                       | -0.20                  | 6.19‡                           | -0.69                               | -0.52      | -1.67‡                     | -1.12** |
| Lagged Secondary Education Expenditure     | 2.15‡                |                       | 1.16                   | 5.84**                          | -0.47                               | 0.16       | -1.27                      | -0.42   |
| Δ Health & Hospital Expenditure            | -0.87                | 0.20                  |                        | 6.39‡                           | -0.50                               | -0.32      | -1.47                      | -0.92   |
| Lagged Health & Hospital Expenditure       | 0.99                 | -1.16                 |                        | 4.68‡                           | -1.62                               | -1.00      | -2.43                      | -1.58   |
| Δ Parks, Rec., & Natural Res. Expenditure  | -7.26**              | -6.19‡                | -6.39‡                 |                                 | -6.89‡                              | -6.71**    | -7.86‡                     | -7.31** |
| Lagged Parks, Rec., & Natural Res. Exp.    | -3.69                | -5.84**               | -4.68‡                 |                                 | -6.31**                             | -5.68‡     | -7.11**                    | -6.27** |

Clark and Miller (2014 working paper):  
 PRED-POS, Sales taxes to fund health and hospital expenditures

**Table 6: Long-Run and Short-Run Effects of Disaggregated Revenues and Expenditures on Growth 1977-2010**  
 (continued)

| Variable                                   | Higher<br>Educ. Exp. | Second.<br>Educ. Exp. | Health &<br>Hosp. Exp. | Parks, Rec., &<br>Nat. Res. Exp | Wel., Unem., Hous.,<br>& Comm. Exp. | Other Exp. | Net Intergov't.<br>Revenue | Surplus |
|--------------------------------------------|----------------------|-----------------------|------------------------|---------------------------------|-------------------------------------|------------|----------------------------|---------|
| Lag Real Person. Inc. Per Capita           | -6.89*               | -6.89*                | -6.89*                 | -6.89*                          | -6.89*                              | -6.89*     | -6.89*                     | -6.89*  |
| Unemployment Rate                          | -0.76*               | -0.76*                | -0.76*                 | -0.76*                          | -0.76*                              | -0.76*     | -0.76*                     | -0.76*  |
| Population Growth Rate                     | -0.56**              | -0.56**               | -0.56**                | -0.56**                         | -0.56**                             | -0.56**    | -0.56**                    | -0.56** |
| Δ Property Tax                             | 0.11                 | -0.96**               | -0.76                  | -7.15‡                          | -0.26                               | -0.44      | 0.71                       | 0.16    |
| Lagged Property Tax.                       | -2.34**              | -0.19                 | -1.35                  | -6.03**                         | 0.27                                | -0.35      | 1.08                       | 0.23    |
| Δ Sales Tax                                | -0.39                | -1.45*                | -1.26                  | -7.65**                         | -0.76                               | -0.93      | 0.21                       | -0.34   |
| Lagged Sales Tax                           | -2.77**              | -0.62                 | -1.78                  | -6.46**                         | -0.15                               | -0.78      | 0.65                       | -0.20   |
| Δ Individual Income Tax                    | 0.17                 | -0.90                 | -0.70                  | -7.09**                         | -0.20                               | -0.38      | 0.77                       | 0.22    |
| Lagged Individual Income Tax               | -1.21                | 0.94                  | -0.22                  | -4.90‡                          | 1.41                                | 0.78       | 2.21**                     | 1.37**  |
| Δ Corporate Income Tax                     | 1.82                 | 0.75                  | 0.94                   | -5.45                           | 1.44                                | 1.27       | 2.42**                     | 1.87    |
| Lagged Corporate Income Tax                | -1.99                | 0.17                  | -0.99                  | -5.68**                         | 0.63                                | 0.01       | 1.43                       | 0.59    |
| Δ Other Revenue                            | 0.76                 | -0.31                 | -0.11                  | -6.50**                         | 0.39                                | 0.21       | 1.36                       | 0.81**  |
| Lagged Other Revenue                       | -2.12**              | 0.03                  | -1.12                  | -5.81**                         | 0.50                                | -0.13      | 1.30‡                      | 0.46‡   |
| Δ Net Utility Expenditure                  | 0.65                 | 1.72**                | 1.52                   | 7.91**                          | 1.02‡                               | 1.20       | 0.05                       | 0.60    |
| Lagged Net Utility Expenditure             | 3.25**               | 1.10                  | 2.26                   | 6.94**                          | 0.63                                | 1.26       | -0.17                      | 0.67    |
| Δ Police, Fire, & Correct. Facilities      | -1.77                | -0.70                 | -0.90                  | 5.49                            | -1.39                               | -1.22      | -2.37**                    | -1.82** |
| Lagged Police, Fire, & Correct. Facilities | 3.09**               | 0.94                  | 2.10                   | 6.78**                          | 0.47                                | 1.10       | -0.33                      | 0.52    |
| Δ Transportation Expenditure               | -1.29                | -0.22                 | -0.42                  | 5.97‡                           | -0.91                               | -0.74      | -1.89                      | -1.34‡  |
| Lagged Transportation Expenditure          | 1.54                 | -0.61                 | 0.55                   | 5.23‡                           | -1.08‡                              | -0.45      | -1.88*                     | -1.04‡  |
| Δ Higher Education Expenditure             |                      | 1.07                  | 0.87                   | 7.26**                          | 0.37                                | 0.55       | -0.60                      | -0.05   |
| Lagged Higher Education Expenditure        |                      | -2.15‡                | -0.99                  | 3.69                            | -2.62**                             | -1.99‡     | -3.42*                     | -2.58** |
| Δ Secondary Education Expenditure.         | -1.07                |                       | -0.20                  | 6.19‡                           | -0.69                               | -0.52      | -1.67‡                     | -1.12** |
| Lagged Secondary Education Expenditure     | 2.15‡                |                       | 1.16                   | 5.84**                          | -0.47                               | 0.16       | -1.27                      | -0.42   |
| Δ Health & Hospital Expenditure            | -0.87                | 0.20                  |                        | 6.39‡                           | -0.50                               | -0.32      | -1.47                      | -0.92   |
| Lagged Health & Hospital Expenditure       | 0.99                 | -1.16                 |                        | 4.68‡                           | -1.62                               | -1.00      | -2.43                      | -1.58   |
| Δ Parks, Rec., & Natural Res. Expenditure  | -7.26**              | -6.19‡                | -6.39‡                 |                                 | -6.89‡                              | -6.71**    | -7.86‡                     | -7.31** |
| Lagged Parks, Rec., & Natural Res. Exp.    | -3.69                | -5.84**               | -4.68‡                 |                                 | -6.31**                             | -5.68‡     | -7.11**                    | -6.27** |

Clark and Miller (2014 working paper):  
 PRED-POS, Sales taxes to fund health and hospital expenditures

**Table 6: Long-Run and Short-Run Effects of Disaggregated Revenues and Expenditures on Growth 1977-2010**  
 (continued)

| Variable                                   | Higher<br>Educ. Exp. | Second.<br>Educ. Exp. | Health &<br>Hosp. Exp. | Parks, Rec., &<br>Nat. Res. Exp | Wel., Unem., Hous.,<br>& Comm. Exp. | Other Exp. | Net Intergov't.<br>Revenue | Surplus |
|--------------------------------------------|----------------------|-----------------------|------------------------|---------------------------------|-------------------------------------|------------|----------------------------|---------|
| Lag Real Person. Inc. Per Capita           | -6.89*               | -6.89*                | -6.89*                 | -6.89*                          | -6.89*                              | -6.89*     | -6.89*                     | -6.89*  |
| Unemployment Rate                          | -0.76*               | -0.76*                | -0.76*                 | -0.76*                          | -0.76*                              | -0.76*     | -0.76*                     | -0.76*  |
| Population Growth Rate                     | -0.56**              | -0.56**               | -0.56**                | -0.56**                         | -0.56**                             | -0.56**    | -0.56**                    | -0.56** |
| Δ Property Tax                             | 0.11                 | -0.96**               | -0.76                  | -7.15‡                          | -0.26                               | -0.44      | 0.71                       | 0.16    |
| Lagged Property Tax                        | -2.34**              | -0.10                 | -1.35                  | -6.03**                         | -0.27                               | -0.35      | 1.08                       | 0.23    |
| Δ Sales Tax                                | -0.39                | -1.45*                | -1.26                  | -7.65**                         | -0.76                               | -0.93      | 0.21                       | -0.34   |
| Lagged Sales Tax                           | -2.77**              | -0.62                 | -1.78                  | -6.46**                         | -0.15                               | -0.78      | 0.65                       | -0.20   |
| Lagged Individual Income Tax               | -1.21                | 0.94                  | -0.22                  | -4.90‡                          | 1.41                                | 0.78       | 2.21**                     | 1.37**  |
| Δ Corporate Income Tax                     | 1.82                 | 0.75                  | 0.94                   | -5.45                           | 1.44                                | 1.27       | 2.42**                     | 1.87    |
| Lagged Corporate Income Tax                | -1.99                | 0.17                  | -0.99                  | -5.68**                         | 0.63                                | 0.01       | 1.43                       | 0.59    |
| Δ Other Revenue                            | 0.76                 | -0.31                 | -0.11                  | -6.50**                         | 0.39                                | 0.21       | 1.36                       | 0.81**  |
| Lagged Other Revenue                       | -2.12**              | 0.03                  | -1.12                  | -5.81**                         | 0.50                                | -0.13      | 1.30‡                      | 0.46‡   |
| Δ Net Utility Expenditure                  | 0.65                 | 1.72**                | 1.52                   | 7.91**                          | 1.02‡                               | 1.20       | 0.05                       | 0.60    |
| Lagged Net Utility Expenditure             | 3.25**               | 1.10                  | 2.26                   | 6.94**                          | 0.63                                | 1.26       | -0.17                      | 0.67    |
| Δ Police, Fire, & Correct. Facilities      | -1.77                | -0.70                 | -0.90                  | 5.49                            | -1.39                               | -1.22      | -2.37**                    | -1.82** |
| Lagged Police, Fire, & Correct. Facilities | 3.09**               | 0.94                  | 2.10                   | 6.78**                          | 0.47                                | 1.10       | -0.33                      | 0.52    |
| Δ Transportation Expenditure               | -1.29                | -0.22                 | -0.42                  | 5.97‡                           | -0.91                               | -0.74      | -1.89                      | -1.34‡  |
| Lagged Transportation Expenditure          | 1.54                 | -0.61                 | 0.55                   | 5.23‡                           | -1.08‡                              | -0.45      | -1.88*                     | -1.04‡  |
| Δ Higher Education Expenditure             |                      | 1.07                  | 0.87                   | 7.26**                          | 0.37                                | 0.55       | -0.60                      | -0.05   |
| Lagged Higher Education Expenditure        |                      | -2.15‡                | -0.99                  | 3.69                            | -2.62**                             | -1.99‡     | -3.42*                     | -2.58** |
| Δ Secondary Education Expenditure          | -1.07                |                       | -0.20                  | 6.19‡                           | -0.69                               | -0.52      | -1.67‡                     | -1.12** |
| Lagged Secondary Education Expenditure     | 2.15‡                |                       | 1.16                   | 5.84**                          | -0.47                               | 0.16       | -1.27                      | -0.42   |
| Δ Health & Hospital Expenditure            | -0.87                | 0.20                  |                        | 6.39‡                           | -0.50                               | -0.32      | -1.47                      | -0.92   |
| Lagged Health & Hospital Expenditure       | 0.99                 | -1.16                 |                        | 4.68‡                           | -1.62                               | -1.00      | -2.43                      | -1.58   |
| Δ Parks, Rec., & Natural Res. Expenditure  | -7.26**              | -6.19‡                | -6.39‡                 |                                 | -6.89‡                              | -6.71**    | -7.86‡                     | -7.31** |
| Lagged Parks, Rec., & Natural Res. Exp.    | -3.69                | -5.84**               | -4.68‡                 |                                 | -6.31**                             | -5.68‡     | -7.11**                    | -6.27** |

Clark and Miller (2014 working paper):  
 PRED-POS, Sales taxes to fund health and hospital expenditures

**Table 6: Long-Run and Short-Run Effects of Disaggregated Revenues and Expenditures on Growth 1977-2010**  
 (continued)

| Variable                                   | Higher<br>Educ. Exp. | Second.<br>Educ. Exp. | Health &<br>Hosp. Exp. | Parks, Rec., &<br>Nat. Res. Exp | Wel., Unem., Hous.,<br>& Comm. Exp. | Other Exp. | Net Intergov't.<br>Revenue | Surplus |
|--------------------------------------------|----------------------|-----------------------|------------------------|---------------------------------|-------------------------------------|------------|----------------------------|---------|
| Lag Real Person. Inc. Per Capita           | -6.89*               | -6.89*                | -6.89*                 | -6.89*                          | -6.89*                              | -6.89*     | -6.89*                     | -6.89*  |
| Unemployment Rate                          | -0.76*               | -0.76*                | -0.76*                 | -0.76*                          | -0.76*                              | -0.76*     | -0.76*                     | -0.76*  |
| Population Growth Rate                     | -0.56**              | -0.56**               | -0.56**                | -0.56**                         | -0.56**                             | -0.56**    | -0.56**                    | -0.56** |
| Δ Property Tax                             | 0.11                 | -0.96**               | -0.76                  | -7.15‡                          | -0.26                               | -0.44      | 0.71                       | 0.16    |
| Lagged Property Tax.                       | -2.34**              | -0.19                 | -1.35                  | -6.03**                         | 0.27                                | -0.35      | 1.08                       | 0.23    |
| Δ Sales Tax                                | -0.39                | -1.45*                | -1.26                  | -7.65**                         | -0.76                               | -0.93      | 0.21                       | -0.34   |
| Lagged Sales Tax                           | -2.77**              | -0.62                 | -1.78                  | -6.46**                         | -0.15                               | -0.78      | 0.65                       | -0.20   |
| Δ Individual Income Tax                    | 0.17                 | -0.90                 | -0.22                  | -7.09**                         | -0.20                               | -0.38      | 0.77                       | 0.22    |
| Lagged Individual Income Tax               | -1.21                | 0.94                  | -0.22                  | -4.90‡                          | 1.41                                | 0.78       | 2.21**                     | 1.37**  |
| Δ Corporate Income Tax                     | 1.82                 | 0.75                  | 0.94                   | -5.45                           | 1.44                                | 1.27       | 2.42**                     | 1.87    |
| Lagged Corporate Income Tax                | -1.99                | 0.17                  | -0.99                  | -5.68**                         | 0.63                                | 0.01       | 1.43                       | 0.59    |
| Δ Other Revenue                            | 0.76                 | -0.31                 | -0.11                  | -6.50**                         | 0.39                                | 0.21       | 1.36                       | 0.81**  |
| Lagged Other Revenue                       | -2.12**              | 0.03                  | -1.12                  | -5.81**                         | 0.50                                | -0.13      | 1.30‡                      | 0.46‡   |
| Δ Net Utility Expenditure                  | 0.65                 | 1.72**                | 1.52                   | 7.91**                          | 1.02‡                               | 1.20       | 0.05                       | 0.60    |
| Lagged Net Utility Expenditure             | 3.25**               | 1.10                  | 2.26                   | 6.94**                          | 0.63                                | 1.26       | -0.17                      | 0.67    |
| Δ Police, Fire, & Correct. Facilities      | -1.77                | -0.70                 | -0.90                  | 5.49                            | -1.39                               | -1.22      | -2.37**                    | -1.82** |
| Lagged Police, Fire, & Correct. Facilities | 3.09**               | 0.94                  | 2.10                   | 6.78**                          | 0.47                                | 1.10       | -0.33                      | 0.52    |
| Δ Transportation Expenditure               | -1.29                | -0.22                 | -0.42                  | 5.97‡                           | -0.91                               | -0.74      | -1.89                      | -1.34‡  |
| Lagged Transportation Expenditure          | 1.54                 | -0.61                 | 0.55                   | 5.23‡                           | -1.08‡                              | -0.45      | -1.88*                     | -1.04‡  |
| Δ Higher Education Expenditure             |                      | 1.07                  | 0.87                   | 7.26**                          | 0.37                                | 0.55       | -0.60                      | -0.05   |
| Lagged Higher Education Expenditure        |                      | -2.15‡                | -0.99                  | 3.69                            | -2.62**                             | -1.99‡     | -3.42*                     | -2.58** |
| Δ Secondary Education Expenditure.         | -1.07                |                       | -0.20                  | 6.19‡                           | -0.69                               | -0.52      | -1.67‡                     | -1.12** |
| Lagged Secondary Education Expenditure     | 2.15‡                |                       | 1.16                   | 5.84**                          | -0.47                               | 0.16       | -1.27                      | -0.42   |
| Δ Health & Hospital Expenditure            | -0.87                | 0.20                  |                        | 6.39‡                           | -0.50                               | -0.32      | -1.47                      | -0.92   |
| Lagged Health & Hospital Expenditure       | 0.99                 | -1.16                 |                        | 4.68‡                           | -1.62                               | -1.00      | -2.43                      | -1.58   |
| Δ Parks, Rec., & Natural Res. Expenditure  | -7.26**              | -6.19‡                | -6.39‡                 |                                 | -6.89‡                              | -6.71**    | -7.86‡                     | -7.31** |
| Lagged Parks, Rec., & Natural Res. Exp.    | -3.69                | -5.84**               | -4.68‡                 |                                 | -6.31**                             | -5.68‡     | -7.11**                    | -6.27** |

## CODING OF TAX VARIABLES (continued)

### ***VARIABLE GROUP = tax measures (Part 4)***

|                       |                                                                                          |
|-----------------------|------------------------------------------------------------------------------------------|
| <i>Pred-Neg</i>       | =1, if the theoretical prediction of the coefficient is negative.                        |
| <i>Pred-Pos</i>       | =1, if the theoretical prediction of the coefficient is positive.                        |
| <i>Pred-Ambiguous</i> | =1, if the theoretical prediction of the coefficient is ambiguous.<br>(Omitted category) |

OTHER STUDY CHARACTERISTICS:

***VARIABLE GROUP = SE measures***

|               |                                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| <i>SE-HAC</i> | =1, if both heteroskedasticity and autocorrelation standard error are considered. |
| <i>SE-HET</i> | =1, if heteroskedasticity standard error is considered.                           |
| <i>SE-OLS</i> | =1, if OLS standard error is considered. (Omitted category)                       |

OTHER STUDY CHARACTERISTICS:

***VARIABLE GROUP = publication characteristics***

|                |                                                                              |
|----------------|------------------------------------------------------------------------------|
| <i>Journal</i> | =1, if the estimate comes from a study published in a peer-reviewed journal. |
| <i>PubYear</i> | The year study was published.                                                |

OTHER STUDY CHARACTERISTICS:

***VARIABLE GROUP = dependent variable (economic growth) measures***

|               |                                                          |
|---------------|----------------------------------------------------------|
| <i>GSP</i>    | =1, if GSP growth is used as DV in the original studies. |
| <i>PC-GSP</i> | =1, if per capita GSP growth is used as DV.              |
| <i>PCPI</i>   | =1, if PCPI growth is used as DV.                        |
| <i>PI</i>     | =1, if PI growth is used as DV. (Omitted category)       |

OTHER STUDY CHARACTERISTICS:

***VARIABLE GROUP = specification characteristics***

|                    |                                                                  |
|--------------------|------------------------------------------------------------------|
| <i>HLK</i>         | =1, if at least 2 of 3 control variables (K, L, H) are included. |
| <i>FE</i>          | =1, if the state fixed effects is considered in estimation.      |
| <i>Initial-Inc</i> | =1, if initial level of output (convergence theory) is included. |
| <i>Lagged-DV</i>   | =1, if lagged dependent variable is included.                    |

OTHER STUDY CHARACTERISTICS:

***VARIABLE GROUP = data characteristics (Part 1)***

|                  |                                                                                           |
|------------------|-------------------------------------------------------------------------------------------|
| <i>Panel</i>     | =1, if panel data as opposed to cross-sectional data is used.                             |
| <i>Interval</i>  | The time interval “between observations” from the same state used in the primary studies. |
| <i>Num-Years</i> | The length of sample time period.                                                         |
| <i>AvgYear</i>   | The average year of data used in the primary studies.                                     |

OTHER STUDY CHARACTERISTICS:

***VARIABLE GROUP = data characteristics (Part 2)***

|                     |                                                                       |
|---------------------|-----------------------------------------------------------------------|
| <i>None</i>         | =1, if all the states are included.                                   |
| <i>AK</i>           | =1, if Alaska is excluded.                                            |
| <i>DC</i>           | =1, if District of Columbia is excluded.                              |
| <i>AKHIDCOthers</i> | =1, if states including (AK, HI, DC, others) are excluded.            |
| <i>AK, HI, DC</i>   | =1, if states including (AK, HI, DC) are excluded. (Omitted category) |

OTHER STUDY CHARACTERISTICS:

***VARIABLE GROUP = estimation characteristics***

|             |                                                         |
|-------------|---------------------------------------------------------|
| <i>GLS</i>  | =1, if Generalized Least Squares estimator is used.     |
| <i>GMM</i>  | =1, if Generalized Method of Moments estimator is used. |
| <i>TSLS</i> | =1, if Two-Stage Least Squares estimator is used.       |
| <i>OLS</i>  | =1, if OLS estimator is used. (Omitted category)        |

## TWO MEASURES OF EFFECT

1) Estimated coefficient (*Bhat*) and its standard error

$$- \hat{\beta}_{tax}$$

$$- se(\hat{\beta}_{tax})$$

2) Partial correlation coefficient (*PCC*) and its standard error

$$- PCC = \sqrt{\frac{t^2}{t^2 + df}}$$

$$- se(PCC) = \sqrt{\frac{1 - PCC^2}{df}}$$

# TESTING FOR PUBLICATION BIAS: The FAT

| <b>Variables</b>              | <b>WLS (FE)</b>  | <b>WLS (RE)</b>     |
|-------------------------------|------------------|---------------------|
|                               | <b>PCC</b>       | <b>PCC</b>          |
| <b>Constant</b>               | 0.002<br>(0.11)  | -0.01<br>(-0.05)    |
| <b>SE</b>                     | -0.96<br>(-1.57) | -0.88***<br>(-2.65) |
| <b>Observations</b>           | 939              | 939                 |
| <b>Adjusted R<sup>2</sup></b> | 0.02             | 0.04                |

## TESTING FOR PUBLICATION BIAS: The FAT

| <b>Variables</b>              | <b>WLS (FE)</b>    | <b>WLS (RE)</b>   |
|-------------------------------|--------------------|-------------------|
|                               | <b>Bhat</b>        | <b>Bhat</b>       |
| <b>Constant</b>               | 0.0003<br>(1.15)   | -0.12<br>(-1.49)  |
| <b>SE</b>                     | -0.89**<br>(-2.59) | -0.45*<br>(-1.74) |
| <b>Observations</b>           | 939                | 939               |
| <b>Adjusted R<sup>2</sup></b> | 0.08               | 0.02              |

# TESTING FOR PUBLICATION BIAS: The FAT

| <i>Variables</i>                     | <i>WLS (FE)</i>  |                    | <i>WLS (RE)</i>     |                   |
|--------------------------------------|------------------|--------------------|---------------------|-------------------|
|                                      | <i>PCC</i>       | <i>Bhat</i>        | <i>PCC</i>          | <i>Bhat</i>       |
| <b><i>Constant</i></b>               | 0.002<br>(0.11)  | 0.0003<br>(1.15)   | -0.01<br>(-0.05)    | -0.12<br>(-1.49)  |
| <b><i>SE</i></b>                     | -0.96<br>(-1.57) | -0.89**<br>(-2.59) | -0.88***<br>(-2.65) | -0.45*<br>(-1.74) |
| <b><i>Observations</i></b>           | 939              | 939                | 939                 | 939               |
| <b><i>Adjusted R<sup>2</sup></i></b> | 0.02             | 0.08               | 0.04                | 0.02              |

COMING UP NEXT:

MULTIPLE META-REGRESSION ANALYSIS  
(Fixed Effects)

| Variables          | Full                 |                       | Stepwise |      |
|--------------------|----------------------|-----------------------|----------|------|
|                    | PCC                  | Bhat                  | PCC      | Bhat |
| <b>SE</b>          | 1.508<br>(1.55)      | -1.166***<br>(-2.87)  |          |      |
| <b>Journal</b>     | -0.006<br>(-0.25)    | -0.084<br>(-0.89)     |          |      |
| <b>PubYear</b>     | -0.0005<br>(-0.22)   | 0.006<br>(1.65)       |          |      |
| <b>GSP</b>         | 0.042<br>(1.32)      | 0.0002<br>(1.51)      |          |      |
| <b>PC-GSP</b>      | 0.078**<br>(1.99)    | 0.246***<br>(2.87)    |          |      |
| <b>PCPI</b>        | 0.083***<br>(2.63)   | 0.093<br>(1.13)       |          |      |
| <b>PI-Tax</b>      | 0.079**<br>(2.41)    | -0.004<br>(-0.73)     |          |      |
| <b>CO-Tax</b>      | 0.092**<br>(2.06)    | -0.004<br>(-0.71)     |          |      |
| <b>SA-Tax</b>      | 0.023<br>(0.63)      | 0.001<br>(0.11)       |          |      |
| <b>PR-Tax</b>      | 0.038<br>(1.24)      | -0.006<br>(-1.18)     |          |      |
| <b>OT-Tax</b>      | -0.033<br>(-1.11)    | -0.006<br>(-1.26)     |          |      |
| <b>Marginal</b>    | -0.081***<br>(-3.76) | 0.004<br>(0.55)       |          |      |
| <b>Differenced</b> | -0.031<br>(-1.41)    | -0.052<br>(-0.96)     |          |      |
| <b>Num-Tax</b>     | 0.002<br>(0.48)      | 0.001***<br>(6.14)    |          |      |
| <b>ETR</b>         | -0.091***<br>(-3.90) | -0.071<br>(-1.50)     |          |      |
| <b>First-Lag</b>   | -0.003<br>(-0.26)    | -0.008***<br>(-6.69)  |          |      |
| <b>More-Lag</b>    | 0.018<br>(1.23)      | -0.008***<br>(-11.33) |          |      |

| Variables                     | Full                 |                      | Stepwise          |                     |
|-------------------------------|----------------------|----------------------|-------------------|---------------------|
|                               | PCC                  | Bhat                 | PCC               | Bhat                |
| <b>Pred-Neg</b>               | -0.005<br>(-0.41)    | 0.012<br>(1.54)      |                   |                     |
| <b>Pred-Pos</b>               | -0.038<br>(-1.50)    | -0.894***<br>(-2.96) |                   |                     |
| <b>HLK</b>                    | 0.0005<br>(0.02)     | -0.015<br>(-0.73)    |                   |                     |
| <b>FE</b>                     | -0.024<br>(-1.44)    | 0.001<br>(0.15)      |                   |                     |
| <b>Initial-Inc</b>            | -0.037<br>(-1.20)    | -0.019<br>(-0.22)    |                   |                     |
| <b>Lag-DV</b>                 | 0.020<br>(0.66)      | -0.011<br>(-0.17)    |                   |                     |
| <b>SE-HAC</b>                 | -0.023<br>(-0.75)    | -0.044<br>(-0.46)    |                   |                     |
| <b>SE-HET</b>                 | -0.038<br>(-0.66)    | -0.114<br>(-1.46)    |                   |                     |
| <b>CSTS</b>                   | -0.015<br>(-0.20)    | -0.116<br>(-1.33)    |                   |                     |
| <b>Interval</b>               | -0.013*<br>(-1.80)   | -0.0001*<br>(-1.79)  |                   |                     |
| <b>Num-Years</b>              | 0.003***<br>(2.54)   | 0.0001<br>(0.58)     |                   |                     |
| <b>AvgYear</b>                | -0.002<br>(-0.90)    | 0.002<br>(0.67)      |                   |                     |
| <b>GLS</b>                    | -0.066***<br>(-2.64) | -0.111<br>(-1.13)    |                   |                     |
| <b>GMM</b>                    | -0.056***<br>(-2.52) | -0.074<br>(-0.46)    |                   |                     |
| <b>TSLS</b>                   | 0.034<br>(1.16)      | 0.085<br>(0.87)      |                   |                     |
| <b>None</b>                   | -0.016<br>(-0.32)    | -0.170**<br>(-2.24)  |                   |                     |
| <b>AK</b>                     | 0.081**<br>(2.42)    | 0.416**<br>(2.66)    |                   |                     |
| <b>DC</b>                     | 0.024<br>(1.18)      | 0.004<br>(0.05)      |                   |                     |
| <b>AKHIDCOthers</b>           | -0.080***<br>(-2.70) | 0.011<br>(0.65)      |                   |                     |
| <b>Constant</b>               | 4.758<br>(1.44)      | -15.970*<br>(-1.96)  | -0.023<br>(-0.78) | -8.176**<br>(-2.74) |
| <b>Observations</b>           | 939                  | 939                  | 939               | 939                 |
| <b>Studies</b>                | 27                   | 27                   | 27                | 27                  |
| <b>Adjusted R<sup>2</sup></b> | 0.240                | 0.174                | 0.223             | 0.170               |

| Variables          | Full                 |                       | Stepwise |      |
|--------------------|----------------------|-----------------------|----------|------|
|                    | PCC                  | Bhat                  | PCC      | Bhat |
| <b>SE</b>          | 1.508<br>(1.55)      | -1.166***<br>(-2.87)  |          |      |
| <b>Journal</b>     | -0.006<br>(-0.25)    | -0.084<br>(-0.89)     |          |      |
| <b>PubYear</b>     | -0.0005<br>(-0.22)   | 0.006<br>(1.65)       |          |      |
| <b>GSP</b>         | 0.042<br>(1.32)      | 0.0002<br>(1.51)      |          |      |
| <b>PC-GSP</b>      | 0.078**<br>(1.99)    | 0.246***<br>(2.87)    |          |      |
| <b>PCPI</b>        | 0.083***<br>(2.63)   | 0.093<br>(1.13)       |          |      |
| <b>PI-Tax</b>      | 0.079**<br>(2.41)    | -0.004<br>(-0.73)     |          |      |
| <b>CO-Tax</b>      | 0.092**<br>(2.06)    | -0.004<br>(-0.71)     |          |      |
| <b>SA-Tax</b>      | 0.023<br>(0.63)      | 0.001<br>(0.11)       |          |      |
| <b>PR-Tax</b>      | 0.038<br>(1.24)      | -0.006<br>(-1.18)     |          |      |
| <b>OT-Tax</b>      | -0.033<br>(-1.11)    | -0.006<br>(-1.26)     |          |      |
| <b>Marginal</b>    | -0.081***<br>(-3.76) | 0.004<br>(0.55)       |          |      |
| <b>Differenced</b> | -0.031<br>(-1.41)    | -0.052<br>(-0.96)     |          |      |
| <b>Num-Tax</b>     | 0.002<br>(0.48)      | 0.001***<br>(6.14)    |          |      |
| <b>ETR</b>         | -0.091***<br>(-3.90) | -0.071<br>(-1.50)     |          |      |
| <b>First-Lag</b>   | -0.003<br>(-0.26)    | -0.008***<br>(-6.69)  |          |      |
| <b>More-Lag</b>    | 0.018<br>(1.23)      | -0.008***<br>(-11.33) |          |      |

| Variables                     | Full                 |                      | Stepwise          |                     |
|-------------------------------|----------------------|----------------------|-------------------|---------------------|
|                               | PCC                  | Bhat                 | PCC               | Bhat                |
| <b>Pred-Neg</b>               | -0.005<br>(-0.41)    | 0.012<br>(1.54)      |                   |                     |
| <b>Pred-Pos</b>               | -0.038<br>(-1.50)    | -0.894***<br>(-2.96) |                   |                     |
| <b>HLK</b>                    | 0.0005<br>(0.02)     | -0.015<br>(-0.73)    |                   |                     |
| <b>FE</b>                     | -0.024<br>(-1.44)    | 0.001<br>(0.15)      |                   |                     |
| <b>Initial-Inc</b>            | -0.037<br>(-1.20)    | -0.019<br>(-0.22)    |                   |                     |
| <b>Lag-DV</b>                 | 0.020<br>(0.66)      | -0.011<br>(-0.17)    |                   |                     |
| <b>SE-HAC</b>                 | -0.023<br>(-0.75)    | -0.044<br>(-0.46)    |                   |                     |
| <b>SE-HET</b>                 | -0.038<br>(-0.66)    | -0.114<br>(-1.46)    |                   |                     |
| <b>CSTS</b>                   | -0.015<br>(-0.20)    | -0.116<br>(-1.33)    |                   |                     |
| <b>Interval</b>               | -0.013*<br>(-1.80)   | -0.0001*<br>(-1.79)  |                   |                     |
| <b>Num-Years</b>              | 0.003***<br>(2.54)   | 0.0001<br>(0.58)     |                   |                     |
| <b>AvgYear</b>                | -0.002<br>(-0.90)    | 0.002<br>(0.67)      |                   |                     |
| <b>GLS</b>                    | -0.066***<br>(-2.64) | -0.111<br>(-1.13)    |                   |                     |
| <b>GMM</b>                    | -0.056***<br>(-2.52) | -0.074<br>(-0.46)    |                   |                     |
| <b>TSLS</b>                   | 0.034<br>(1.16)      | 0.085<br>(0.87)      |                   |                     |
| <b>None</b>                   | -0.016<br>(-0.32)    | -0.170**<br>(-2.24)  |                   |                     |
| <b>AK</b>                     | 0.081**<br>(2.42)    | 0.416**<br>(2.66)    |                   |                     |
| <b>DC</b>                     | 0.024<br>(1.18)      | 0.004<br>(0.05)      |                   |                     |
| <b>AKHIDCOthers</b>           | -0.080***<br>(-2.70) | 0.011<br>(0.65)      |                   |                     |
| <b>Constant</b>               | 4.758<br>(1.44)      | -15.970*<br>(-1.96)  | -0.023<br>(-0.78) | -8.176**<br>(-2.74) |
| <b>Observations</b>           | 939                  | 939                  | 939               | 939                 |
| <b>Studies</b>                | 27                   | 27                   | 27                | 27                  |
| <b>Adjusted R<sup>2</sup></b> | 0.240                | 0.174                | 0.223             | 0.170               |

| Variables          | Full                 |                       | Stepwise             |                       |
|--------------------|----------------------|-----------------------|----------------------|-----------------------|
|                    | PCC                  | Bhat                  | PCC                  | Bhat                  |
| <b>SE</b>          | 1.508<br>(1.55)      | -1.166***<br>(-2.87)  | 1.142<br>(1.38)      | -0.972***<br>(-2.87)  |
| <b>Journal</b>     | -0.006<br>(-0.25)    | -0.084<br>(-0.89)     | -                    | -0.074**<br>(-2.47)   |
| <b>PubYear</b>     | -0.0005<br>(-0.22)   | 0.006<br>(1.65)       | -                    | 0.003***<br>(3.29)    |
| <b>GSP</b>         | 0.042<br>(1.32)      | 0.0002<br>(1.51)      | -                    | -                     |
| <b>PC-GSP</b>      | 0.078**<br>(1.99)    | 0.246***<br>(2.87)    | 0.035***<br>(2.58)   | 0.204***<br>(3.18)    |
| <b>PCPI</b>        | 0.083***<br>(2.63)   | 0.093<br>(1.13)       | -                    | 0.088***<br>(3.57)    |
| <b>PI-Tax</b>      | 0.079**<br>(2.41)    | -0.004<br>(-0.73)     | 0.081***<br>(7.20)   | -0.0002***<br>(-2.81) |
| <b>CO-Tax</b>      | 0.092**<br>(2.06)    | -0.004<br>(-0.71)     | 0.093***<br>(3.46)   | -0.0002**<br>(-2.35)  |
| <b>SA-Tax</b>      | 0.023<br>(0.63)      | 0.001<br>(0.11)       | -                    | -                     |
| <b>PR-Tax</b>      | 0.038<br>(1.24)      | -0.006<br>(-1.18)     | 0.038***<br>(4.72)   | -                     |
| <b>OT-Tax</b>      | -0.033<br>(-1.11)    | -0.006<br>(-1.26)     | -                    | -                     |
| <b>Marginal</b>    | -0.081***<br>(-3.76) | 0.004<br>(0.55)       | -0.088***<br>(-5.49) | -                     |
| <b>Differenced</b> | -0.031<br>(-1.41)    | -0.052<br>(-0.96)     | -                    | -                     |
| <b>Num-Tax</b>     | 0.002<br>(0.48)      | 0.001***<br>(6.14)    | -                    | 0.001**<br>(2.78)     |
| <b>ETR</b>         | -0.091***<br>(-3.90) | -0.071<br>(-1.50)     | -0.077***<br>(-5.96) | -0.075***<br>(-3.46)  |
| <b>First-Lag</b>   | -0.003<br>(-0.26)    | -0.008***<br>(-6.69)  | -                    | -0.009***<br>(-3.91)  |
| <b>More-Lag</b>    | 0.018<br>(1.23)      | -0.008***<br>(-11.33) | -                    | -0.008***<br>(-5.94)  |

| Variables                     | Full                 |                      | Stepwise             |                      |
|-------------------------------|----------------------|----------------------|----------------------|----------------------|
|                               | PCC                  | Bhat                 | PCC                  | Bhat                 |
| <b>Pred-Neg</b>               | -0.005<br>(-0.41)    | 0.012<br>(1.54)      | -                    | -                    |
| <b>Pred-Pos</b>               | -0.038<br>(-1.50)    | -0.894***<br>(-2.96) | -                    | -0.947***<br>(-3.04) |
| <b>HLK</b>                    | 0.0005<br>(0.02)     | -0.015<br>(-0.73)    | -                    | -                    |
| <b>FE</b>                     | -0.024<br>(-1.44)    | 0.001<br>(0.15)      | -                    | -                    |
| <b>Initial-Inc</b>            | -0.037<br>(-1.20)    | -0.019<br>(-0.22)    | -                    | -                    |
| <b>Lag-DV</b>                 | 0.020<br>(0.66)      | -0.011<br>(-0.17)    | -                    | -                    |
| <b>SE-HAC</b>                 | -0.023<br>(-0.75)    | -0.044<br>(-0.46)    | -                    | -                    |
| <b>SE-HET</b>                 | -0.038<br>(-0.66)    | -0.114<br>(-1.46)    | -                    | -0.065**<br>(-2.35)  |
| <b>CSTS</b>                   | -0.015<br>(-0.20)    | -0.116<br>(-1.33)    | -                    | -                    |
| <b>Interval</b>               | -0.013*<br>(-1.80)   | -0.0001*<br>(-1.79)  | -0.011*<br>(-1.83)   | -                    |
| <b>Num-Years</b>              | 0.003***<br>(2.54)   | 0.0001<br>(0.58)     | 0.003***<br>(4.14)   | -                    |
| <b>AvgYear</b>                | -0.002<br>(-0.90)    | 0.002<br>(0.67)      | -                    | 0.001*<br>(2.05)     |
| <b>GLS</b>                    | -0.066***<br>(-2.64) | -0.111<br>(-1.13)    | -0.049**<br>(-2.21)  | -0.108***<br>(-2.86) |
| <b>GMM</b>                    | -0.056***<br>(-2.52) | -0.074<br>(-0.46)    | -0.046***<br>(-4.10) | -0.125***<br>(-3.43) |
| <b>TSLS</b>                   | 0.034<br>(1.16)      | 0.085<br>(0.87)      | -                    | -                    |
| <b>None</b>                   | -0.016<br>(-0.32)    | -0.170**<br>(-2.24)  | -                    | -0.122***<br>(-4.42) |
| <b>AK</b>                     | 0.081**<br>(2.42)    | 0.416**<br>(2.66)    | -                    | 0.354***<br>(3.38)   |
| <b>DC</b>                     | 0.024<br>(1.18)      | 0.004<br>(0.05)      | -                    | -                    |
| <b>AKHIDCOthers</b>           | -0.080***<br>(-2.70) | 0.011<br>(0.65)      | -0.085**<br>(-2.42)  | -                    |
| <b>Constant</b>               | 4.758<br>(1.44)      | -15.970*<br>(-1.96)  | -0.023<br>(-0.78)    | -8.176**<br>(-2.74)  |
| <b>Observations</b>           | 939                  | 939                  | 939                  | 939                  |
| <b>Studies</b>                | 27                   | 27                   | 27                   | 27                   |
| <b>Adjusted R<sup>2</sup></b> | 0.240                | 0.174                | 0.223                | 0.170                |

AND NOW:

MULTIPLE META-REGRESSION ANALYSIS  
(Random Effects)

| Variables          | Full                 |                      | Stepwise             |                      |
|--------------------|----------------------|----------------------|----------------------|----------------------|
|                    | PCC                  | Bhat                 | PCC                  | Bhat                 |
| <b>SE</b>          | 1.252*<br>(1.85)     | -0.347<br>(-1.15)    | 0.564<br>(1.53)      | -0.479*<br>(-1.91)   |
| <b>Journal</b>     | 0.0003<br>(0.01)     | 0.083<br>(0.51)      | -                    | -                    |
| <b>PubYear</b>     | -0.001<br>(-0.79)    | -0.010<br>(-0.93)    | -                    | -                    |
| <b>GSP</b>         | 0.024<br>(0.86)      | 0.077<br>(0.46)      | -                    | -                    |
| <b>PC-GSP</b>      | 0.049<br>(1.62)      | 0.370<br>(1.53)      | -                    | -                    |
| <b>PCPI</b>        | 0.054**<br>(2.32)    | 0.306<br>(1.57)      | -                    | -                    |
| <b>PI-Tax</b>      | 0.072***<br>(2.89)   | 0.159<br>(1.15)      | 0.077***<br>(3.72)   | 0.301**<br>(2.20)    |
| <b>CO-Tax</b>      | 0.105***<br>(3.34)   | 0.430<br>(1.29)      | 0.108***<br>(4.36)   | 0.569*<br>(2.05)     |
| <b>SA-Tax</b>      | 0.015<br>(0.58)      | -0.333*<br>(-1.96)   | -                    | -0.190**<br>(-2.48)  |
| <b>PR-Tax</b>      | 0.040**<br>(2.03)    | -0.260<br>(-1.58)    | 0.045***<br>(3.87)   | -                    |
| <b>OT-Tax</b>      | -0.050*<br>(-1.85)   | -0.480**<br>(-2.79)  | -0.038***<br>(-2.77) | -0.331***<br>(-3.96) |
| <b>Marginal</b>    | -0.109***<br>(-2.69) | -0.473**<br>(-2.44)  | -0.115***<br>(-3.42) | -0.500***<br>(-3.43) |
| <b>Differenced</b> | -0.043**<br>(-2.13)  | -0.218<br>(-1.62)    | -0.054***<br>(-3.22) | -0.227*<br>(-1.85)   |
| <b>Num-Tax</b>     | -0.002<br>(-0.42)    | 0.039<br>(0.96)      | -                    | 0.058*<br>(1.94)     |
| <b>ETR</b>         | -0.071***<br>(-2.96) | -0.487***<br>(-3.44) | -0.083***<br>(-4.94) | -0.316***<br>(-3.17) |
| <b>First-Lag</b>   | 0.003<br>(0.30)      | 0.031<br>(0.31)      | -                    | -                    |
| <b>More-Lag</b>    | 0.028*<br>(1.75)     | 0.117<br>(0.91)      | -                    | -                    |
| <b>Pred-Neg</b>    | -0.008<br>(-0.49)    | -0.073<br>(-0.59)    | -                    | -                    |
| <b>Pred-Pos</b>    | -0.049*<br>(-1.84)   | -0.880**<br>(-2.70)  | -                    | -0.782**<br>(-2.46)  |

| Variables           | Full                 |                     | Stepwise             |                      |
|---------------------|----------------------|---------------------|----------------------|----------------------|
|                     | PCC                  | Bhat                | PCC                  | Bhat                 |
| <b>HLK</b>          | -0.043**<br>(-2.38)  | -0.144<br>(-1.19)   | -0.041**<br>(-2.44)  | -0.236***<br>(-3.84) |
| <b>FE</b>           | -0.046***<br>(-2.62) | -0.313*<br>(-1.88)  | -0.049***<br>(-2.87) | -0.174*<br>(-1.74)   |
| <b>Initial-Inc</b>  | -0.052**<br>(-2.11)  | -0.069<br>(-0.33)   | -0.041**<br>(-2.32)  | -                    |
| <b>Lag-DV</b>       | 0.049<br>(1.60)      | 0.192<br>(0.78)     | 0.078***<br>(3.11)   | -                    |
| <b>SE-HAC</b>       | -0.033<br>(-0.83)    | 0.003<br>(0.02)     | -                    | -                    |
| <b>SE-HET</b>       | -0.059<br>(-1.41)    | -0.044<br>(-0.26)   | -                    | -                    |
| <b>CSTS</b>         | 0.103**<br>(2.20)    | -0.080<br>(-0.27)   | 0.134**<br>(2.45)    | -                    |
| <b>Interval</b>     | -0.007<br>(-1.37)    | -0.013<br>(-0.81)   | -                    | -                    |
| <b>Num-Years</b>    | 0.002***<br>(2.64)   | -0.006<br>(-1.29)   | -                    | -                    |
| <b>AvgYear</b>      | -0.0002<br>(-0.10)   | 0.009<br>(0.84)     | -                    | -                    |
| <b>GLS</b>          | -0.060***<br>(-2.84) | -0.332<br>(-1.69)   | -0.069***<br>(-2.68) | -                    |
| <b>GMM</b>          | -0.048<br>(-1.26)    | -0.637*<br>(-1.72)  | -0.063***<br>(-3.95) | -0.525**<br>(-2.11)  |
| <b>TSLS</b>         | 0.035<br>(1.11)      | -0.72<br>(-0.52)    | -                    | -                    |
| <b>None</b>         | -0.023<br>(-0.60)    | -0.291<br>(-1.13)   | -                    | -                    |
| <b>AK</b>           | 0.108***<br>(2.66)   | 0.533<br>(1.61)     | 0.112***<br>(4.40)   | -                    |
| <b>DC</b>           | 0.042*<br>(1.82)     | -0.302*<br>(-1.92)  | -                    | -                    |
| <b>AKHIDCOthers</b> | -0.057*<br>(-1.90)   | -0.818**<br>(-2.22) | -                    | -                    |
| <b>Constant</b>     | 2.466<br>(0.84)      | 2.637<br>(0.14)     | -0.088<br>(-1.28)    | 0.239**<br>(2.28)    |
| <b>Observations</b> | 939                  | 939                 | 939                  | 939                  |
| <b>Studies</b>      | 27                   | 27                  | 27                   | 27                   |

## SUMMARY:

Different conclusions from multiple MRA depending on whether one uses *PCC* or *Bhat*

## QUESTIONS FOR OTHER RESEARCHERS:

- 1) Has anybody noticed this before?
- 2) Which set of results are more dependable (*PCC* or *Bhat*)?